Expander Graph and it's Applications

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Declaration

I, Soham Sarkar hereby declare that the matter of this project report entitled *Expander Graph and it's Applications* is record of the survey/research work carried out by me, at the Department of Mathematics, Savitribai Phule Pune Univerity, Pune under the supervision of Prof. Ganesh S. Kadu, Department of Mathematics, Savitribai Phule Pune University in partially fulfilling the requirements of the course MT-38 : Spectral Graph Theory

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Abstract

In this project, we survey *expander graphs* in a very succinct manner. In the first chapter, we give an introduction to different ways of defining *expander graphs*. In the second chapter, we discuss some properties of *expander graphs*. In the third chapter, we focus on algebraic, analytic and combinatorial approaches to construct *expander graphs*. Finally, in the last chapter, we discuss some applications of *expander graphs* and proved the Expander-mixing lemma, in particular.

Contents

Chapter 1

Introduction

The term *expander graph* was first coined by Pinsker in 1973, in the field of computer science (error correction codes, communication networks and algorithms). Informally, an *expander graph* is a highly connected, sparse(there are relatively few edges), finite, undirected multigraph in which, every subset of vertices that is not too large has a boundary.

So, every connected graph is an expander; however, different connected graphs have different expansion parameters. Intuitively, a graph is a good *expander* if each of it's vertices have a low degree and high expansion parameters.

1.1 Definition

Different formalisations of the above informal introduction gives rise to different notions of *expander graphs*, namely in terms of edge expansion, vertex expansion and spectral expansion.

1.1.1 Edge expansion (Edge isoperimetric number/Edge expansion ratio/Cheeger constant)

Let X = (V,E) be a k-regular graph on n vertices, where V is the set of vertices and E is the edge set of X.
Then, the **edge expansion ratio** of X is defined as follows :
h(X):= $\min_{0<|S|\leq\frac{n}{2}} \frac{|\partial S|}{|S|}$,
where S $\subseteq$ V and E(S,X$\setminus$S) or ∂S:= $\{$$\{u,v\}$ $\in E(X) : u \in S, v \in V(X)\setminus$S$\}$.

The graph X is an ε- *expander graph* if $\exists$ an $\varepsilon > 0$, such that the expansion ration h(X) $> \varepsilon$.

1.1.2 Vertex Expansion (Vertex isoperimetric number)

Let X be the graph as stated above and S $\subseteq$ V. Then the **vertex isoperimetric numbers** h_{out}(X) and h_{in}(X) for the above graph X are :

h_{out}(X):= $\min_{0<|S|\leq\frac{n}{2}}\frac{|\partial_{out}(S)|}{|S|}$ and h_{in}(X):= $\min_{0<|S|\leq\frac{n}{2}}\frac{|\partial_{in}(S)|}{|S|}$,
where $\partial_{out}(S)$ is the outer boundary of S i.e. the set of vertices in V(X)$\diagdown$S with atleast one neigbour in S; and
$\partial_{in}(S)$ is the inner boundary of S i.e. the set of vertices of S with atleast one neighbour in V(X)$\diagdown$S.

1.1.3 Spectral Expansion

If A is an adjacency matrix of the k-regular graph X on n vertices, say, then, by Spectral theorem, A has n real eigen values $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n$ in [-k,k].

Thus, the distribution(uniform distribution because X is regular) u = $[\frac{1}{n}, \frac{1}{n}, ..., \frac{1}{n}]^t$ $\in$R^n is a stationary distribution of X i.e. Au = ku, u($\neq$ 0) is an eigen vector of A with eigen value λ_1 = k. The spectral gap of X is defined by k - λ_2, which measures the **spectral expansion** of X.

Let $\varepsilon > 0$ and k $\geq$ 1. Then, the finite k-regular graph X is a *one-sided ε-expander* if one has $\lambda_2 \leq$ (1 - ε)k and a *two-sided ε-expander* if one also has $\lambda_n \geq$ - (1 - ε)k. A family {X_i} of such finite k-regular graphs is a *one-sided expander* family if $\exists$ $\varepsilon > 0$, such that each X_i is a *one-sided ε-expander.*

The operator Δ = 1 - $\frac{1}{k}$A is the Graph Laplacian, where
A: l^2(V) $\rightarrow$l^2(V) the adjacency operator on functions f: X $\rightarrow \mathbb{C}$ defined by Af(v) = $\sum_{\{w\in V:\{v,w\}\in E\}}$f(w). Then, this operator Δ is a positive semidefinite operator with atleast one zero eigen value (corresponding to eigen vector 1). A graph is an *ε-expander* if and only if $\exists$ a spectral gap of size ε in Δ in the sense that the first eigen value of Δ exceeds the second by atleast ε.

Chapter 2

Properties

A more precise relation between the expansion ratio h and the previously stated ε, which makes the same graph X a one-sided *ε-expander* is the discrete Cheeger inequality :
$\frac{\varepsilon}{2}k \leq$ h(X) $\leq \sqrt{2\varepsilon}$k (First proved by Dodzuik, Alon-Milman).

Some properties of *expander graphs* are stated below :

2.1 Expanders have low diameter

Let X = (V,E) be a k-regular, one-sided *ε-expander graph* on n-vertices for some n > k $\geq$ 1 and $\varepsilon > 0$. Then, $\exists$ a constant c > 0 depending only on k and ε, such that $\forall$ v $\in$ V and any radius r $\geq$ 0, the ball $B(v, r)$ = {w $\in$ V : d(v,w) < r} has cardinality $|B(v, r)| \geq$ min{(1+c)^r, n} for some metric d on X.

The bound on the diameter of X is given by diam(X) << log(3|V|), which depend on the expansion parameter and the number of vertices of X.

2.2 Expanders have high connectivity

Let a k-regular graph X on n-vertices be a one-sided ε- *expander* for some n > k $\geq$ 1 and $\varepsilon > 0$. If one removes m edges from X, then, the resulting graph has a connected component of size atleast (n - Cm), where C depends only on k and ε.

2.3 Expanders have a high chromatic number

Let a k-regular graph X on n-vertices be a one-sided ε- *expander* for some n $>$ k $\geq$ 1 and $\varepsilon > 0$. Then, any independent set in X has cardinality atmost (1 - ε)n. So, the chromatic number of X is atleast $\frac{1}{1-\varepsilon}$ as a consequence of Expander Mixing Lemma discussed later in chapter-4. This ε is still useful for constructing bounded-degree graphs of high chromatic number and large girth although as ε comes close to 0, the chromatic number becomes non-trivial.

2.4 Expanders and their concentration of measure

Let a k-regular graph X on n-vertices be a one-sided ε-*expander* for some n $>$ k $\geq$ 1 and $\varepsilon > 0$. Let f: X $\rightarrow \mathbb{R}$ be a Lipschitz function with Lipschitz constant K. Then, |f(v) - f(w)| $\leq$ K$d(v,w)$. Let M be a median value of f (i.e. f(v) $\geq$ M for atleast half of the vertices v and f(v) $<$ M for the other half atleast).

Then, $|\{v \in V : |f(v) - M| \geq \lambda K\}| \leq$ Cne$^{-c\lambda}$, $\forall\ \lambda > 0$ and some constants C, c $>$ 0 depending only on K, ε.

Chapter 3

Construction of Expanders

There are three general strategies for constructing families of *expander graphs* - first strategy being algebraic and group-theoretic, second being analytic and additive combinatorial approach while the third is via zig-zag graph products.
Few methods have been described here.

3.1 Discrete Torus expanders

Let X = (V,E) be a graph with vertex set V = $\mathbb{Z}_M \times \mathbb{Z}_M$ and edges from each node (x,y) to the nodes (x,y), (x+1,y), (x,y+1), (x,x+y), (-y,x) under arithmetic modulo M.

3.2 p-cycle inverse chords

Let X = (V,E) be a graph with V = $\mathbb{Z}_p$ and edges that connect each node x with nodes : x+1, x-1 and x^{-1} under arithmetic modulo p(0^{-1} is defined to be 0).

3.3 Ramanujan Graphs

Let X = (V,E) be a graph with V = $\mathbb{F} \cup \{\infty\}$, the finite field of prime order q, such that q $\equiv$ 1(mod 4) plus one extra node for infinity. The edges in this graph connect each node z with all such z′ of the form :
$z' = \frac{(a_0+ia_1)z+(a_2+ia_3)}{(-a_2+ia_3)z+(a_0-ia_1)}$,
where $a_0, a_1, a_2, a_3 \in \mathbb{Z}$, such that $a_0^2 + a_1^2 + a_2^2 + a_3^2 =$ p, a_0 is odd and positive, a_1, a_2, a_3 are even for some fixed p $\neq$ q, such that p $\equiv$ 1(mod

4), q is square modulo p and i $\in \mathbb{F}_q$, such that $i^2 \equiv -1 \pmod q$.

3.4 Zig-zag product

Let G be a D_1-regular digraph on N_1 vertices and H a D_2-regular digraph on D_1 vertices. Then, G$\ominus$H is a graph whose vertices are pairs of the form $(u,i) \in [N_1]\times[D_1]$.

For a, b $\in [D_2]$, the (a,b)-th neighbour of a vertex (u,i) is the vertex (v,j) computed as follows :

- i' be the a^{th} neighbour of i in H.
- v be the i^{th} neighbour of u in G, so, e = (u,v) is the i'-th edge leaving u. Let j' be such that e is the j'-th edge entering v in G.
- j be the b^{th} neighbour of j' in H.

3.5 Mildly explicit Expanders

Let H be a $(D^4, D, \frac{7}{8})$-graph and define $G_1 = H^2$ and $G_{t+1} = G_t^2 \ominus H$. This gives a family of *expanders* $\{G_i\}$.

3.6 Fully explicit Expanders

Let H be a $(D^8, D, \frac{7}{8})$-graph and define $G_1 = H^2$ and
$G_{t+1} = (G_t \otimes G_t)^2 \ominus H$.

Chapter 4

The Applications

Expander graphs have applications in various fields in complexity theory, theory of building robust computer networks, theory of error-correcting codes with efficient encoding and decoding, etc. An *expander graph* with bounded valence is an asymptotic graph with the number of edges growing linearly with respect to the number of vertices.

4.1 Expander-Mixing Lemma

Let X = (V,E) be a connected, k-regular, non-bipartite graph on n vertices and $\mu = \max\{\ |\lambda| : \lambda \in \text{spec(A)}\ ,\ |\lambda| \neq \text{k}\}$.
Then, $|E(S,T) - \frac{k|S||T|}{n}| \leq \mu\sqrt{|S||T|}$, where $E(S,T)$ denotes the edges of X between S and T, such that S, T $\subset$ V.
i.e. for the above graph X with edge density $\frac{k}{n}$ (where n = $|V|$), the value of $E(S,T)$ will be closer to the expected number of edges between S and T if the second largest eigen value μ(in absolute value) of the adjacency matrix A of X is as small as possible.

Proof : By our hypothesis, $\mu <$ k.

Let δ_S denote the vector whose entries are 1 for the vertices of S and 0 otherwise. The Spectral theorem for the symmetric matrix A, the adjacency matrix of X states that there exists a complete orthonormal basis of $\mathbb{R}^n$ consisting of eigen vectors ϕ_j of A_j , i.e., for which,

$\text{A}\phi_j = \lambda_j\phi_j$ and $(\text{A})_{ab} = \sum_{j=1}^{n} \lambda_j\phi_j(a)\phi_j(b)$,
where $\phi_j(a)$ denotes the entry of ϕ_j corresponding to a vertex a of X. We assume a numbering such that $\phi_1(a) = \frac{1}{\sqrt{n}}$ and $\lambda_1 =$ k. Then, pulling out the first term of the sum gives
$|E(S,T)| = \delta_S^t\text{A}\delta_T = \sum_{j=1}^{n} \lambda_j \sum_{a\in S, b\in T} \phi_j(a)\phi_j(b)$

$\Rightarrow |E(S,T)| = \frac{k}{n}|S||T| + \sum_{j=2}^{n} \lambda_j \sum_{a\in S, b\in T} \phi_j(a)\phi_j(b)$
Now, by the definition of μ, since our graph is not bipartite, there is only one eigen value with absolute value equal to k, and
$|\sum_{j=2}^{n} \lambda_j \sum_{a\in S, b\in T} \phi_j(a)\phi_j(b)| \le \mu \sum_{j=2}^{n} \lambda_j \sum_{a\in S, b\in T} |\phi_j(a)\phi_j(b)|$.
To finish the proof, use Cauchy-Schwarz inequality. Note that the Fourier coefficients of δ_S with respect to the basis ϕ_j are
$\langle \delta_S, \phi_j \rangle = \sum_{a\in S} \phi_j(a)$.
This implies by Bessel's equality that
$|S| = \|\delta_S\|_2^2 = \sum_{j=1}^{n} \langle \delta_S, \phi_j \rangle^2$.
So, Cauchy-Schwarz inequality gives
$|\sum_{j=2}^{n} \lambda_j \sum_{a\in S, b\in T} \phi_j(a)\phi_j(b)|$
$\le |\sum_{j=1}^{n} \lambda_j \sum_{a\in S, b\in T} \phi_j(a)\phi_j(b)| \le \sqrt{|S||T|}$.
The proof of the lemma follows.

4.2 Barzdin-Kolmogorov Graph Embedding Theorem

Let Γ be a finite graph with degree/valency atmost 6 at each vertex. Then,

- Let t be a thick embedding of Γ. $\exists$ a constant c $>$ 0 independent of Γ and t, such that radius rad(t) $\ge$ c$\sqrt{h(\Gamma)|V|}$.
- $\exists$ a constant c $>$ 0 independent of Γ, such that Γ admits a thick embedding with radius $\le$ c$\sqrt{|V|}$

4.3 Ellenberg-Hall-Kowalski

Let X_Γ be a compact, connected Riemann surface of genus g $\ge$ 2 and S be a finite symmetric set of Γ. For n $\ge$ 1, let $\Gamma_n \lhd \Gamma$(a normal subgroup) with finite index, such that [Γ:Γ_n] $\to \infty$ as n $\to \infty$. If the family of relative Cayley graphs $(\text{C}(\frac{\Gamma}{\Gamma_n}, S))_{n\ge1}$ is an expander family, then, $\gamma(X_\Gamma) \to \infty$ as n $\to \infty$. In fact, then, $\gamma(X_{\Gamma_n})$ $>>$ [Γ:Γ_n], where the implied constants depend on S and the expansion parameters of the family.

4.4 Expander walk sampling

Chernoff bound states that when many independent samples from a random variable in the range [-1,1] is with high probability, then, the average of the samples is close to the expectation of the random variable. The Sampling lemma(due to Aj Tai, Komlós and Szemerédi(1987)) states that this is true when sampling from a walk on an *expander graph*.

4.5 Expander property of Braingraphs

MRI data shown by Szalkai, et al. describes that the graph of the anatomical brain is a significantly better *expander* in women than in men.

Chapter 5

Conclusion

On one hand, an easy application of the probabilistic method shows that a randomly chosen graph will be an *expander* is of very high probability. On the other hand, one wants an *expander*, which is more deterministic in nature. For applications in number theory or geometric group theory, is to determine the expansion properties of symmetric graphs like Cayley graph or more general Schreier graph. Such questions are related to deep properties of various groups of Lie type, such as Kazhdan's property.

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